Supplement information 2. The data characteristics derived from the selected studies.

First author, year of publication	Studied SNPs	Country	Genotyping method	Sample size (case/control)	P_{HWE}
Abigail W. Bushley,2004	IL-18 rs187238, IL-10 rs1800871	America	ABI 7900 sequence detection system	182/219,181/219	0.99/<0.001
Adrianna Mostowska,2013	Bsm1 rs1544410, Fok1 rs2228570	Poland	PCR-RFLP	168/182,168/182	0.94/0.06
Adrianna Mostowska,2016	Bsm1 rs1544410, Fok1 rs2228570	Poland	PCR-RFLP	245/465,245/465	0.35/0.04
Agnieszka Honorata Ludwig,2009	PGR rs10895068	Poland	TaqMan	251/352	0.94
Alamtaj Samsami Dehaghani,2009	IL-18 rs187238	Iron	PCR	85/158	0.69
Amal M.H. Mackawy,2019	XRCC3 rs861539, XRCC3 rs1799794	Egypt	PCR/RFLP	50/20,50/19	0.47/0.8
Amanda B.Spurdle,2001	EPHX1 rs1051740, PGR rs1042839, PGR rs1042838	Australia	ABI Prism 7700	545/287,551/289, 551/289	0.31/0.17/ 0.17
Andrea Romano,2006	PGR rs1042839, PGR rs10895068	Netherland	PCR-RFLP	67/433,58/379	0.12/0.94
Andrew Berchuck,2004	PGR rs10895068	America Australia	TaqMan	438/504 535/298	0.94 0.94
Anıl Çağla	MTHFR rs1801133	Turkry	PCR-RFLP	50/54	0.63

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Anna Jakubowska,2007	MTHFR rs1801133, MTHFR rs1801131, RAD51 rs1801320	Poland	PCR-RFLP	144/280,146/280, 127/127	0.03/0.1/ 0.17
Anna Jakubowska,2010	ERCC2 rs1799793, ERCC2 rs13181, COMT rs4680, PGR rs10895068	Poland	PCR-RFLP	144/280,145/280, 144/280,144/279	0.59/0.42/ 0.53/0.93
Annika Auranen,2005	BRCA1 rs1799950, BRCA1 rs799917, NBS1 rs1063045, NBS1 rs1805794, NBS1 rs709816, NBS1 rs1061302, RAD51 rs1801320, RAD51 rs1801321, RAD52 rs11226, XRCC2 rs3218536, XRCC3 rs1799794, XRCC3 rs1799796, XRCC3 rs861539	UK(SEARCH)	TaqMan	1349/2021,1640/2920,	0.18/0.29/
		US(FROC)	PCR	1661/3528,1586/2792,	0.63/0.46/
		Danish(MALOVA)	PCR	1409/3899,1561/3743,	0.62/0.40/
				1629/2805,1644/2602,	0.82/0.38/
				1649/39061588/3618,	0.80/0.04/
		UK(RMH/YOV)	PCR	1658/3900,1664/3966	0.09/0.41/
				1670/3849	0.75
BEATA SMOLARZ,2013	RAD51 rs1801320, RAD51 rs1801321	Poland	PCR-RFLP	210/210,210/210	0.54/0.51
Beata Smolarz,2019	XRCC2 rs718282, RAD51 rs1801320	Poland	PCR-RFLP	600/600,600/600	0.57/0.55
Catherine M. Phelan,2010	GALNT1 rs17647532	America，England，Australia，Polish，Danish	TaqMan	7794/9314	0.11
D. V. Khokhrin,2012	ERCC2 rs1799793, ERCC2 rs13181, XRCC1 rs25487, GSTP1 rs1695	Russia	PCR	104/298,104/298, 104/298,104/298,	0.66/0.12/ 0.98/0.39

Dan Tong,2001	PGR rs1042839, PGR rs1042838	Australia	PCR	226/194,226/194	0.14/0.14
Daniela B. Leite,2008	PGR rs1042838	Brazil	PCR	182/80	0.21
Delores J. Grant,2013	Apa1 rs7975232, Taq1 rs731236, Bsm1 rs1544410, VDR rs2239179, VDR rs3782905, LINC02354 rs7968585	America	TaqMan	706/765,585/609, 572/601580/607, 581/605,582/609	0.41/0.85/ 0.4/0.78/ 0.46/0.2
Dominique Bernard-Gallon,2008	ERCC2 rs1799793, ERCC2 rs13181	France	TaqMan	51/994,52/995	0.67/0.34
E. Cecchin, 2004	CYP1B1 rs1056836	Italy	PCR-RFLP	220/280	0.45
Ece Konac,2007	VEGF rs833061, VEGF rs3025039	Turkry	PCR	47/104,47/104	0.4/0.2
Elena Ioana Braicu,2006	IL-10 rs1800896, IL-10 rs1800871	Germany	PCR	147/129,197/129	0.46/0.77
Elizabeth I. O. Garner. 2002	COMT rs4680	America	PCR-RFLP	210/225	0.39
Ellen L. Goode,2011	EPHX1 rs1051740	America,Australia	Illumina GoldenGate assay	924/1036	0.01
Emina J. Malisic,2015	RAD51 rs1801320, XRCC1 rs25487	Serbia	PCR-RFLP	50/78,50/78	0.91/0.52
Faten Zahran Mohamed,2013	ERCC2 rs13181, XRCC2 rs3218536, CDKN1B rs2066827	Egypt	PCR	100/100,100/100, 100/100	0.28/0.46/ 0.45
Galina Lurie,2007	APAI rs7975232, BsmI rs1544410, TaqI rs731236, FOk1 rs2228570, Cdx-2 rs11568820	America	PCR-RFLP	163/307,566/601, 161/319,164/316, 162/316	<0.01/<0.01/ <0.01/0.66 /<0.01
Galina Lurie,2009	ESR2 rs3020450	America	TaqMan	313/569	<0.0001

Galina Lurie,2010	PTGS2 rs5275	America	TaqMan	1025/1687	0.04
Galina Lurie,2011	FOK1 rs2228570	Denmark，England，America	TaqMan	1820/3479	0.01
Haifeng Qiu,2017	HOTAIR rs920778	China	TaqMan	190/380,139/300	0.03
Haijing Wu,2016	LncRNA-HOTAIR rs4759314	China	RT-PCR	1000/1000	0.01
Hanna Romanowicz,2016	ERCC2 rs238406	America	PCR	400/100	0.01
Hanna Romanowicz,2017	ERCC2 rs238406	Poland	PCR	400/400	0.54
Harvey A. Risch,2006	PGR rs10895068	America	PCR	487/533	0.96
Hoenil Jo,2007	ERCC1 rs11615	Koera	PCR-RFLP	94/329	0.71
Honglin Song,2006	RB1 rs4151551, RB1 rs3092904, RB1 rs4151636	Denmark，England，America	TaqMan	1486/4310,1332/4191, 1481/4761	0.99/0.88/ 0.27
Honglin Song,2009	H19 rs2107425	Denmark,England,America，Australia	TaqMan	5366/8538	0.86
Houda Bouanene,2011	MUC16 rs2547065	Tunisia	PCR	117/76	0.41
Ian Harley,2008	MLH1 rs1800734	Canada	MassARRAY	310/576	<0.01
János Lukács,2019	miR-146a rs2910164，miR-196a-2 rs11614913	Hungary	PCR	75/75, 75/75	0.09/ 0.38
Jennifer A.	ESR1/SYNE1 rs2295190	America	TaqMan	486/657	0.87

Doherty,2010		America		294/671	0.19
		America		348/538	0.21
		Australia		876/1100	0.06
		Germany		203/427	0.37
		America		69/158	0.37
		Denmark		439/794	0.71
		America		274/366	0.30
		America		610/738	0.62
		America		605/901	0.31
		Poland		254/589	0.88
		England		937/1213	0.35
		America		266/181	1.00
		America		278/420	0.06
		England		468/563	0.05
Jin X,2008	CDKN1B rs2066827	China	PCR-RFLP	234/284	0.34
Johnathan M. Lancaster,1996	EPHX1 rs1051740	America	PCR-RFLP	73/75	0.37
Johnathan M. Lancaster,2003	PGR rs1042838	America	PCR-RFLP	309/397	0.04
Jonathan Beesley,2007	SRD5A2 rs523349, HSD17B4 rs17145454, HSD17B1 rs605059, CYP19A1 rs10046, BRCA2 rs144848, XRCC2 rs3218536, XRCC3 rs861539	Australia	MALDI-TOF	1442/1791, 1418/1774, 1354/1750 1438/1796,1422/1773, 1409/17871235/1674	0.69/ 0.08/ 0.44/ 0.47/0.26/ 0.92/0.62

José Augusto Rinck-Junior,2015	VEGF rs3025039	Brazil	PCR	131/137	0.87
Julie E. Goodman. 2000	COMT rs4680	Germany	PCR-RFLP	108/106	0.86
Karolina Tecza,2015	PGR rs1042838, TP53 rs1042522, GSTP1 rs1695	Poland	ASA-PCR，multiplex-PCR，RFLP-PCR	244/345,225/341, 224/334	0.34/0.93/ 0.30
Kathryn L. Terry,2005	PGR rs10895068，PGR rs1042839，PGR rs1042838	America	TaqMan	920/960,896/939, 896/939	0.95/0.19/ 0.19
Kathryn L. Terry,2010	MTHFR rs1801131, MTHFR rs1801133, GALNT2 rs2271077	America，England	TaqMan	1555/1989,1576/2019,2413/2946	0.302/0.145/ 0.110
Kristina A. Williams,2014	MUC16 rs2547065	America	TaqMan	699/732	0.37
L. YAN,2008	p16/CDKN2 rs11515, p16/CDKN2 rs3088440	China	PCR-RFLP	205/208,205/208	0.05/0.88
Laetitia Delort,2008	CYP1B1 rs1056836, CYP1B1 rs1800440, PGR rs10895068, COMT rs4680	France	TaqMan	51/1000,51/1000,51/1000, 51/1000	0.43/ 0.19/0.92/ 0.48
Leilei Niu,2015	hMLH1 rs1800734	China	MALDI-TOF	421/689	<0.001
Li Li,2015	MCP-1 rs1024611	China	PCR-RFLP	275/293	0.56
Li Yan,2013	Fas rs2234767	China	AS-PCR	342/344	0.68
Li Zhang,2012	MTHFR rs1801133	China	TaqMan	215/218	0.02
Lydia Quaye,2009	H19 rs2107425, XRCC3 rs861539, CDKN2A rs11515, CDKN2A rs3088440	Denmark，England America	TaqMan	1457/2463,1332/2024,1479/2492,1475/2490	<0.01/0.62/ 0.15/0.91

Lyudmila F. Gulyaeva,2008	CYP1A2 rs762551	Russia	PCR-RFLP	96/180	<0.01
M.S. Monteiro,2014	ERCC2 rs1799793, XRCC3 rs861539	Brazil	PCR-RFLP	70/70,70/70	0.02/0.37
Magdalena M. Michalska,2014	XRCC2 rs718282	Poland	PCR-RFLP	608/400	0.48
Magdalena M. Michalska,2016	XRCC2 rs3218536, XRCC3 rs861539	Poland	PCR-RFLP	700/700,700/700	0,54/0.46
Marc T. Goodman, 2001	CYP1B1 rs1056836, COMT rs4680	America	PCR-RFLP	128/144,125/144	0.41/0.21
Marc T. Goodman,2003	CYP1A2 rs762551	America	PCR-RFLP	164/194	0.29
Masatsugu Ueda,2009	MDM2 rs2279744	Tokyo	PCR-RFLP	85/108	0.04
Merete Bjørnslett,2012	MDM2 rs117039649, MDM2 rs2279744	Norway	PCR	1566/2465,2566/2465	0.96/0.67
Mingyao Zhang,2020	APEX1 rs1130409, APEX1 rs1760944	China	TaqMan，PCR	193/231,192/272	0.80/0.41
Narmella Saeedi,2020	LncRNA-HOTAIR rs4759314	Iron	PCR-RFLP	100/100	0.01
Ni, J, and Huang, Y,2016	miR-196a2 rs11614913	China	PCR-RFLP	75/75	0.46
NJ McKenna,1995	PGR rs1042838	Ireland	PCR-RFLP	67/184	0.12
Nora Alyahri,2019	BRCA1 rs799917	Saudi Arabia	TaqMan	28/92	0.98
Penelope M.	RAD51 rs1801320, XRCC2 rs3218536,	Australia	PCR-RFLP	997/2097,954/2068,	0.002/0.164/

Webb,2005	XRCC3 rs861539			991/2077	0.245
Piotr Pawlik,2011	MTHFR rs1801133, MTR rs1805087, MTRR rs1801394	Poland	PCR-RFLP	135/160,136/160, 134/160	0.12/0.17/ 0.97
PM Webb,2011	MTHFR rs1801131, MTR rs1805087, MTRR rs1801394	Australia	MassARRAY	1638/1278,1638/1278,1638/1228,1638/1277	0.8/0.3/ 0.8/1.0
R. Attar,2017	MCP1 rs1024611	Turkry	PCR-RFLP	56/52	0.15
Rachel T. Palmieri,2008	IL18 rs1834481	Australia,America,England	Illumina	6925/8589	0.32
Rafał Watrowski,2015	HER2 rs1136201	Austria	PCR	142/100	0.53
Richard A. DiCioccio,2004	STK15 rs2273535, STK15 rs1047972, STK15 rs732417, STK15 rs8173	England	TaqMan	752/843,750/843, 750/840,717/812	0.54/0.51/ 0.95/0.14
		America	PCR	308/398,323/427, 319/413,299/401	0.42/0.98/ 0.59/0.50
		Denmark	PCR	334/723,432/1112, 390/949,385/963	0.20/0.96/0.34/0.05
Romanowicz-Makowska H,2012	RAD51 rs1801320	Poland	PCR-RFLP	120/120	0.07
S. Dholariya,2016	TP53 rs1042522	New Delhi	ASO-PCR	70/70	0.09
S.W. Baxter,2002	EPHX1 rs1051740	England	AS-PCR	291/257	0.30
Sandra Costa,2007	ERCC2 rs1799793, ERCC2 rs238406	Portugal	PCR-RFLP	123/199,188/187	0.74/0.49
Santhanam Shanmughapriya,2013	HER2 rs1136201	India	PCR–RFLP	72/288	0.09

Sarah K. Holt, 2007	CYP1B1 rs1056836, CYP1B1rs1056827, CYP1B1 rs1800440, COMT rs4680	America	TaqMan	310/574,310/576, 310/577,310/576	0.01/0.19/ 0.38/0.08
Shan Kang,2004	EPHX1 rs1051740	China	PCR-RFLP	86/174	<0.001
Shan Kang,2008	MDM2 rs2279744	China	PCR	257/257	0.45
Shan-Yang He,2012	ERCC1 rs11615	China	TaqMan	155/312	0.77
Shelley S. Tworoger,2009	Bsm1 rs1544410, Fok1 rs2228570, Cdx2 rs11568820	America	TaqMan	1391/1885,1392/1894, 1405/1910	0.11/0.06/ 0.02
Simon A. Gayther,2007	CDKN2A rs3731249, CDKN2A rs11515, CDKN1B rs2066827	England	TaqMan	1478/2452,1053/1276, 1495/2484	0.39/0.38/ 0.004
Simone P. Pinheiro,2010	PPAR-γ/PPARG rs1801282, COX-2 rs20417, COX-2 rs5275	America	TaqMan	224/644,1075/1136, 224/645,1050/1111, 217/612,1046/1103	0.12/ 0.82/0.34
Song CX,2012	MTHFR rs1801131	China	PCR	200/200	0.29
Song Gao,2012	MTHFR rs1801133	China	PCR-RFLP	224/432	0.26
Stian Knappskog,2011	MDM2 rs117039649	Norway，Netherland	PCR	832/2518	0.96
Susanne Schüler,2014	ESR2 rs3020450	Poland，Germany	PCR	182/184	0.65
Tadeusz De˛bniak,2006	CDKN2A rs3731249	Poland	PCR	340/3000	0.02
Tess V. Clendenen,2008	BSMI rs1544410, APAI rs7975232, TAQI rs731236, FOK1 rs2228570	America,Sweden	pyrosequencing	168/318,168/320, 169/320,168/321	0.2/0.37/ 0.07/0.94

Thomas A. Sellers, 2005	CYP1B1 rs1056836, CYP1B1 rs1056827, COMT rs4680	America	PCR	490/598,490/596,489/596	0.45/ 0.28/0.5
Thomas A. Sellers,2008	GALNT1 rs17647532, FUT3 rs2306969, GALNT6 rs907352, GALNT7 rs934358, MGAT5 rs1257187, ST3GAL3 rs3828139, ST3GAL3 rs37460	America	PCR	385/461, 444/478,829/939, 827/939,829/940, 829/940, 828/941	<0.001/ 0.02/0.02/ 0.01/0.02/ 0.04/ 0.05
TP Manolitsas,1997	PGR rs1042838	England	PCR-RFLP	231/220	0.14
UZAY GORMUS,2007	Fas rs2234767, FasL rs763110	Turkry	PCR	47/41,47/41	0.27/0.68
Vidudala V.T.S. Prasad,2011	MTHFR rs1801133	India	PCR-RFLP	80/125	0.96
Wendy M. Smith,2001	PPARG rs1801282	England	PCR	31/62	0.61
		Tokyo	PCR	28/215	1.00
		America	PCR	26/80	0.43
X.C. Sun,2016	miR-146a rs2910164, miR-196a2 rs11614913	China	PCR	134/277, 134/277	0.37/ 0.36
Xiaohong Zhang,2013	APE1 rs1760944, APE1 rs1130409	China	PCR	124/141,124/141	0.56/0.87
Xiaoyan Liu,2015	miR-146a rs2910164, miR-196a2 rs11614913	China	PCR-RFLP	75/100, 75/100	<0.01/ 0.44
Xin Wei,2015	MCP-1 rs1024611	China	PCR–RFLP	257/273	0.58
Yajing Feng,2019	ESR2 rs3020450	China	TaqMan	680/680	0.71

Yuan C,2015	Fok1 rs2228570	China	PCR	129/298	0.43
Yuxia Bao,2020	ERCC-1 rs11615, ERCC-1 rs3212986, ERCC-1 rs2298881	China	PCR	599/323, 599/323,599/323	0.23/ 0.2/0.24
Zahra Mojtahedi,2013	HER2 rs1136201	Iron	PCR–RFLP	107/130	0.14
Zhiguang Zhao,2018	ERCC1 rs3212986, ERCC1 rs11615, ERCC1 rs2298881	China	Real-time PCR	89/356, 89/356,89/256	0.840/ 0.895/0.9
ZHI-SHUANG SONG,2016	miR-196a2 rs11614913	China	PCR	479/431	0.44
Yali Hao,2012	FASL rs763110	China	PCR-RFLP	342/344	0.74
Xiuping He,2008	IL-10 rs1800896, IL-10 rs1800871	China	PCR	33/90,33/90	0.11/0.31
Jinghui Jia,2009	VEGF rs3025039	China	PCR-RFLP	256/329	0.86
Pengfei Liu,2007	XRCC2 rs718282	China	PCR-RFLP	220/244	0.91
Yang Ruan,2014	MCP-1 rs1024611	China	PCR-RFLP	92/38	0.62
Wengang Si,2019	hMLH1 rs1800734	China	PCR-RFLP	536/529	0.21
Yan Wang,2010	VEGF rs833061	China	PCR-RFLP	303/303	0.21
Yan Wu,2007	MTHFR rs1801133	China	PCR	81/86	0.02
Yanping Xing,2007	XPD rs1799793	China	PCR-RFLP	235/246	0.92
Anqi Zhang,2019	ERCC1 rs3212986, ERCC1 rs11615, ERCC1 rs2298881	China	TaqMan	192/272, 193/271,192/272	0.167/ 0.111/0.840